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Date: 1 August 2025

Solution to GATE EC 2018 - Question 31

Solution:

A four-variable Boolean function is realized using two 4×1 multiplexers.

Determining the minimized boolean expression for the output function $F(U, V, W, X)$:-

Considering the first MUX

Inputs to the first (left) MUX:

$$I_0 = 0, \quad I_1 = 1, \quad I_2 = 1, \quad I_3 = 0$$

Select lines: $S_1 = U, \quad S_0 = V$

So the output A of the first MUX is determined by:

U	V	Selection	A
0	0	$I_0 = 0$	0
0	1	$I_1 = 1$	1
1	0	$I_2 = 1$	1
1	1	$I_3 = 0$	0

From the truth table, we identify: $A = U\overline{V} + \overline{U}V$

which is the XOR operation, i.e., $A = U \oplus V$

Considering the second MUX

Inputs to the second (right) MUX:

$$I_0 = A\overline{W}, \quad I_1 = A\overline{w}, \quad I_2 = 0, \quad I_3 = 0$$

Select lines: $S_1 = W, \quad S_0 = X$

So the output F of the second MUX is:

W	X	Selection	F
0	0	$I_0 = A\overline{W}$	$A\overline{W}$
0	1	$I_1 = A\overline{W}$	$A\overline{W}$
1	0	$I_2 = 0$	0
1	1	$I_3 = 0$	0

So, $F = A\overline{W} + A\overline{W}$
Now substitute $A = U\overline{V} + \overline{U}V$

$$F = (U\overline{V} + \overline{U}V)(\overline{W})$$

Final Answer

$$\boxed{F(U, V, W, X) = (U\overline{V} + \overline{U}V)(\overline{W})}$$

This corresponds to **Option (C)**.